

Research Proposal

Design-Based Sensitivity Analysis for Matched Observational Studies under Hidden Bias and Model-Generated Measurements

1. Summary

Matched observational studies are the standard design for estimating treatment effects when randomization is impossible, and their credibility rests on one object: the distribution of treatment assignments within matched sets. Classical practice takes that distribution to be uniform and asks, through a sensitivity analysis, how far it could depart from uniformity before a conclusion is overturned [1]. My current work with Prof. Xinran Li [2] shows that the anchor should instead be the fitted propensity odds, that departures should be bounded on the odds scale by a parameter Λ , and that a lower-envelope statistic gives studentized tests of Neyman's weak null that are valid over the whole band of admissible laws. This proposal generalizes that principle, an **anchor**, a **bounded departure**, and a **guarantee over the band**, to the two ways in which the data of a modern matched study depart from the ideal: the assignment law may depend on unmeasured covariates (**hidden bias**); and covariates and outcomes are increasingly produced by language models from text rather than measured (**model-generated measurements**). Each departure is treated as a perturbation of a law, anchored at an estimate, bounded by an interpretable parameter, and carried through randomization inference rather than assumed away. The two aims are (1) design sensitivity and optimal contrasts for covariate-adaptive full matching; (2) design-based inference with model-generated covariates and outcomes and a small human-verified calibration sample. Each aim delivers theory, open-source software, and an application in clinical or science-of-science data.

2. Background and preliminary results

2.1 Setting. There are K matched sets; set k contains n_k units with treatment indicators Z_{kj} , potential outcomes $Y_{kj}(1), Y_{kj}(0)$, and observed outcome $Y_{kj} = Z_{kj}Y_{kj}(1) + (1 - Z_{kj})Y_{kj}(0)$. The target is Neyman's one-sided weak null $H_0(\tau_0) : \bar{\tau}_K \leq \tau_0$ on the matched-sample average effect, which restricts only the average and imputes no missing potential outcome. Randomization inference for weak nulls requires studentization to remain valid under heterogeneous effects [3, 4]; Rosenbaum's model allows the treatment odds of two units in a set to differ by at most Γ and reports the largest Γ at which a conclusion survives [1]. Inexact matching leaves residual propensity discrepancies that invalidate the uniform reference law [5], and covariate-adaptive randomization inference replaces it with probabilities derived from fitted propensity odds [6].

2.2 My results. In [2] we combine the two. With baseline odds r_{k1}, r_{k2} in pair k , the anchor is $\theta_k = r_{k1}/(r_{k1} + r_{k2})$; a unit-level hidden-bias bound Γ_u induces, after conditioning on one treated unit per pair, the pair-level parameter $\Lambda = \Gamma_u^2$ and the interval $q_k \in [L_k(\Lambda), U_k(\Lambda)]$ with $L_k = \theta_k/\{\theta_k + (1 - \theta_k)\Lambda\}$ and $U_k = \theta_k/\{\theta_k + (1 - \theta_k)/\Lambda\}$. The proposed statistic is an asymmetric Horvitz–Thompson lower envelope: a positive observed contrast is divided by the largest feasible probability of observing it and a negative one by the smallest. We prove that the envelope score is mean-controlled and pointwise sharp within a zero-normalized observed-cell class, that it reduces to the known-propensity Horvitz–Thompson statistic at $\Lambda = 1$, and that its studentized average gives asymptotically valid tests, with a randomization-calibrated critical value, an honest inflated-endpoint version for estimated propensities, and a validity theorem for the plug-in test under fast propensity estimation. The construction extends to full matching [7] in closed form: a level-free difference contrast depends on the assignment vector only through the realized special unit's coordinate, whose marginal range is again a scalar interval, so the entire theory carries over. Within full matching,

the simple within-set mean is one member of a family of balanced, pretreatment matrix-weighted contrasts W that share every guarantee, and a propensity-anchored local member recovers substantial power in large heterogeneous sets. In the right heart catheterization study the covariate-adaptive analysis reports a sensitivity value of $\Lambda = 1.40$ against 1.30 for the uniform analysis. Two features of this work carry into the proposal: the anchor is a function of estimated propensities, so any error in the covariates displaces it (Aim 2), and the choice of W and of the matched structure is a free design decision whose consequences for power are not yet understood (Aim 1).

2.3 Related work and gap. Design sensitivity [8] and split-sample design choice [9] exist only for uniform reference laws and sharp nulls, and average-case sensitivity analysis [10] offers a less conservative alternative to worst-case bounds; none of these has been developed for the covariate-adaptive anchor or for weak nulls in full matching. Prediction-powered inference [11–13] and design-based supervised learning [14] correct model-generated labels with a human-labelled subsample, but for population means and regressions, not for matched designs, sensitivity analysis, or the covariate side. That is the gap this proposal fills.

3. Aim 1: Design sensitivity and optimal contrasts for covariate-adaptive full matching

Problem. Our theory shows that in full matching a whole family of weighted contrasts is valid, but says nothing about which member to use or which matched structure (pairs, fixed ratio, full matching with bounded set sizes) to build. Rosenbaum’s design sensitivity $\tilde{\Gamma}$, the limiting sensitivity parameter up to which a study can reject as $K \rightarrow \infty$ under a favourable model with a real effect and no hidden bias, is the natural criterion [8], but it has been derived only for uniform reference laws.

Approach. Under a favourable model $Y(0) = h(\text{logit } e(X)) + \varepsilon$, $Y(1) = Y(0) + \tau$, I will derive the design sensitivity of the covariate-adaptive studentized weak-null test as a function of the set-size distribution and of W , and characterize the maximizing W ; the conjecture, supported by our simulations, is that the propensity-anchored local weight is optimal whenever propensity locality is prognostically informative. I will then quantify how anchoring itself changes design sensitivity relative to the uniform law when residual discrepancy is prognostic, which turns the empirical finding in the RHC study into a theorem. Because h and the error scale must be estimated, and estimation requires outcomes, I will follow the split-sample protocol of [9], choosing W and the structure on a planning sample and testing on the remainder, and study the optimal planning fraction when the design is chosen by search over many candidates. Recent work on model-assisted randomization tests that avoid sample splitting by fitting once [15] suggests a route to recovering the lost sample, which I will pursue. The computational deliverable is a solver for the envelope optimization over general set structures.

4. Aim 2: Matched studies with model-generated covariates and outcomes

Setting. Each unit has structured covariates X , a text document T (a clinical note, an abstract), and possibly a confounder U recorded only in the text. A language model produces an extraction $\hat{U} = f(T)$ or a predicted outcome $\hat{Y} = g(T)$ for every unit; a calibration subsample $\mathcal{C}$, drawn with known inclusion probabilities π_k , receives human-coded values. Practice has moved faster than theory: in most applications the model output is used as if measured without error.

Outcomes. Let $\hat{D}_k$ be the within-set lower-envelope score computed from $\hat{Y}$ and D_k the same score from Y . The rectified statistic $\hat{\tau}_{PP} = \sum_k \hat{D}_k + \sum_{k \in \mathcal{C}} \pi_k^{-1} (D_k - \hat{D}_k)$ has, conditional on the assignment, the same expectation as the fully labelled statistic *whatever* the model’s error structure, even when the error depends on treatment through the text. Hence its worst case over Λ is attained at the same extremal law and the envelope machinery applies unchanged; validity never rests on the model being accurate. What is new is the variance, which has an assignment component and a

labelling component. I will derive a conservative estimator for the sum, prove asymptotic validity of the studentized weak-null test for every Λ , show that with a tuned weight on the prediction term [12] the test is never less powerful than the test on $\mathcal{C}$ alone, and study the allocation of a fixed labelling budget across sets, for instance toward sets where the model’s predictions are least stable across prompts. When no human labels can be obtained I will develop a (Λ, δ) analysis in which δ bounds the within-set difference in mean model error between treated and control units.

Covariates. Matching and propensity fitting on $(X, \hat{U})$ leaves residual imbalance in U and displaces the anchor. If the logit propensity is Lipschitz in U with constant L , the within-set log odds ratio departs from the anchor by at most $L(|U_{kj} - \hat{U}_{kj}| + |U_{kl} - \hat{U}_{kl}|)$. The extraction error is observed on $\mathcal{C}$ and L is estimated there, so a high quantile of this bound is estimable, giving an extraction-induced parameter Λ_{ext} that, unlike the classical Λ , is identified, and that combines multiplicatively with it on the odds scale. The report is then a function of the unidentified Λ with Λ_{ext} estimated and its sampling uncertainty carried through the studentized test; the technical core is the lower envelope with an anchor that has estimation error of its own, and the target is validity when $|\mathcal{C}|$ grows more slowly than K . Two existing ideas enter here: conditional inference via exchangeable sampling, which calibrates the test conditional on the calibration sample by resampling extraction errors exchangeably [16, 17], and catalytic priors, which stabilize the propensity fit in small calibration samples with synthetic data without sacrificing validity [18]. Post-treatment leakage, where the text mentions treatment, will be audited by comparing propensity fits from time-stamped pre-treatment text with fits from the full text.

Abstention. My work on scientific world models showed that a predictor should return *unsupported* rather than a number when its evidence is seed-noisy or context-dependent, and that doing so raised accuracy on the reported cases from 71% to 100%. I will formalize the analogue here: when calibration diagnostics show that the model’s error is unstable across prompts or strata, the analysis abstains on the affected sets, with error control for the reported conclusions in the spirit of conformal and selective inference [19]. Recent work on conditional evaluation of language models with cheap auxiliary signals [20] is directly relevant.

5. Evaluation and applications

Semi-synthetic studies will attach real text and real model outputs to simulated assignments with controlled hidden bias, so that Λ , Λ_{ext} , and the labelling error are known, and will check size and power across extraction quality, calibration size, and set structure. The right heart catheterization data, which contain no text, serve as the check that every method reduces to [2] when no model-generated measurements are involved. Two applications are planned. The first is clinical: MIMIC-IV and its notes, with severity and comorbidity extracted from admission notes as model-generated covariates, and an ICU treatment decision in the structured data as the exposure. The second is science of science, extending the Knowledge Lab study I contributed to: paper novelty labelled by a language model from abstracts with a human-coded calibration set, used to estimate how exposure to an AI breakthrough changes the direction of a scientist’s subsequent work, subject to the relevant data agreements.

6. Timeline and expected contributions

Year 1: coursework and qualifying examinations; Aim 1 theory and the envelope solver; the semi-synthetic testbed. **Year 2:** Aim 1 split-sample planning; first software release; Aim 2 theory for model-generated outcomes. **Year 3:** Aim 2 theory for model-generated covariates, the abstention framework, and the clinical text application. **Year 4:** science-of-science application, integration, and thesis. Each aim yields self-contained papers, and Aim 2 does not depend on Aim 1. The proposal will deliver design-sensitivity theory and optimal contrasts for covariate-adaptive full matching;

design-based tests for model-generated outcomes and covariates whose validity does not rest on model accuracy; a principled abstention rule; and open-source software in R and Python.

7. Fit with HKU Business School

I propose to carry out this programme in the Innovation and Information Management area of HKU Business School, under the supervision of **Prof. Junting Duan** and **Prof. Xinghao Qiao**. Prof. Duan's work on inference with AI-generated covariates and on imputation-powered inference for missing covariates [21, 22] asks, for estimation problems, the question that Aim 2 asks for matched designs and sensitivity analysis: how to keep inference valid when some variables are produced by a model rather than measured. Carrying that approach into the design-based envelope of Aim 2 is a natural first joint project. Prof. Qiao's current research on causal inference and experimental design, with applications to online A/B testing, and on how LLM-generated data can improve statistical learning, meets the proposal at two points: Aim 1 is itself a design problem, choosing the matched structure and contrast that make a study most robust to hidden bias, and Aim 2 asks when model-generated measurements can enter an analysis without compromising its validity. Every aim also rests on nuisance quantities estimated from many covariates, such as the propensity anchor and the extraction model of Aim 2, whose inputs include text embeddings and longitudinal clinical records. The validity theorems therefore need rates for high-dimensional estimators under dependence, where his work on high-dimensional and functional time series [23, 24] bears directly on the proposal. It also suggests an extension I would like to pursue: matched designs in which each unit's pre-treatment covariates are curves, such as vital-sign trajectories before an ICU decision, so that the anchor is estimated from functional data. The same problems arise in business settings, for example when the effect of a business decision is estimated from observational data, or when outcomes are coded from customer text by a language model, and I would be glad to develop applications of this kind within the School.

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